FamilyForge — pedigree-aware interpretation of a family callset
(annotated VCF + pedigree + panel -> ranked candidates + clinician / publication / clinical reports)

Inputs (provided per family; never hardcoded)

Annotated joint VCF.gz
CallForge stage-11 output
(VEP CSQ · ClinVar · gnomAD · PhyloP)

Gene panel TSV
gene · tier · inheritance
mechanism · omim · notes

Config YAML (optional)
thresholds + gene rules
(overrides default.yaml)

cnv_calls.tsv (optional)
CallForge stage-8 CNV calls

extra-scores TSV (optional)
CADD · REVEL · AlphaMissense
for finalist variants

Pedigree (6-col PED)
family · iid · father · mother
sex · phenotype

provenance.json (optional)
upstream tool / DB versions

override

Ingest & parse (inputs.py · vcf_io.py)

_normalise_vcf (cli.py)
bcftools norm -m-any
(split multiallelics)

load_pedigree / load_panel /
load_config / load_cnv_calls /
load_extra_scores (inputs.py)
-> Family · panel · cfg

load_variants (vcf_io.py)
pick_consequence · AF · ClinVar · PhyloP
per-sample GT/DP/AD/GQ + quality gate
-> Variant objects (panel-restricted)

cnv rows · panel

Family · panel

finalist scores

methods

Interpretation modules

cnv.py
cnv_candidates:
deletions overlapping panel genes
(priority genes e.g. DOCK8)

inheritance.py
evaluate_variant:
de-novo/dominant · AR-hom · X-linked
evaluate_compound_het: AR comp-het (trans)
MZ-twin concordance · segregation strings

inheritance.relatedness_sex_qc
pairwise genotype concordance
+ chrX-het sex inference

CNV tiered directly

candidates

acmg.py
tier_candidate:
ClinVar · AF · consequence · in-silico
conservation · gene mechanism · segregation
-> Tier A / B / C + evidence trail

qc

Reporting (report.py · clinical_report.py · templates/*.j2)

report.build_context
rank_candidates · _var_row · classify
-> ranked rows + tier buckets + QC

clinical_report.py
letterhead engine (data-driven)
lab_config + report_config + candidates.json

report.render (Jinja2)
clinician + publication templates

candidates.json + PED

Outputs (outdir/)

clinician_report.{html,md}
concise · actionable

publication_report.{html,md}
full methods · QC · evidence · limits

candidates.json
machine-readable rows + QC

clinical_report.{html,md}
letterhead PDF via headless Chrome

familyforge (CLI entry point, cli.py:main)
orchestrates the whole run

familyforge-clinical-report
(clinical_report.py:main)

Module groups

- Inputs (per family)
- Ingest / parse
- Inheritance models
- CNV cross-check
- ACMG-style tiering
- Genotype QC
- Reporting
- Outputs
- Console entry point